MUGUNDAN S

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EXPERIENCE

AArch64 Backend Contributor / *LLVM Open-Source Project*

- Extended the **AArch64 cross-block ConditionOptimizer** to handle select-family instructions, eliminating redundant comparisons across control-flow paths [PR#208369].
- Added **DAGCombine support** for vector power-of-two mask comparisons, lowering them to efficient CMTST + BSP instruction sequences [PR#209100].
- Added **DAGCombine support** to transform SUBS with negative immediate comparisons into CMN instructions [PR#221398].
- Added **intra-block Compare+Branch optimization** to generate TBZ/TBNZ from zero-comparison patterns, improving conditional branching efficiency [PR#217808].
- Added **diamond-arm instruction folding** into chained CSEL-family instructions, reducing redundant instructions and improving generated AArch64 code [PR#212033].

Software Engineer | *MulticoreWare, India* **July 2025 – Present**

- Contributed to **90+ open-source projects** for enabling native compilation support on Windows ARM64 devices **[Link]**.
- Developed **CI/CD workflows** for automated build, testing, packaging, and release of projects for Windows ARM64**.**
- Worked on debugging and resolving **hard crashes** related to **JIT generated codes** on Windows ARM64.
- Resolved architecture-specific build and toolchain issues across multiple projects improving Windows ARM64 reliability and ecosystem adoption.
- Improved **ARM64** application performance using **ARM NEON SIMD** optimizations and **WPR/WPA profiling** to identify and eliminate runtime bottlenecks.

Junior Software Engineer – Intern | *MulticoreWare, India* **Mar 2024 – June 2025**

- Ported and enabled **AI models** on **TDA4VM hardware**, validating output consistency against host platforms to ensure deployment accuracy.
- Performed **layer-by-layer model debugging** and accuracy analysis to identify numerical mismatches and resolve inference quality issues on **TDA4VM hardware**.
- Benchmarked and validated inference performance across model pipelines on TDA4VM, identifying regressions and ensuring performance parity after feature integration.

HIGHLIGHTS

Core-Collective – Python Ecosystem for Windows ARM64

- Core member of the Core Collective team - Collaborated with engineers across multiple organizations to stabilize the Python ecosystem for Windows on ARM **[Link]**..
- Contributed **20+ upstream PRs** during the challenge, enabling **Windows ARM64 wheel builds**, CI/CD automation, and package compatibility improvements across the Python ecosystem.

TOOLS & SKILLS

- **Languages & Concepts:** C, C++, Python, Java, OOP
- **Tools & Platforms:** Git, MSYS2, GNU, Microsoft Visual Studio, LLVM, Inno Setup, CMake, Make
- **Specialized Knowledge:** Performance optimization, Computer Architecture, AArch64 Assembly, ARMv8 ISA (Programmers model, Exception modeling and Memory model), Windows on ARM, Embedded Systems, Performance Profiling (WPR, WPA, Profile Explorer).
- **LLVM Compiler:** Code Generation, Instruction Selection (SelectionDAG), AAPCS64(Calling Convention), Fixup & Relocations (MC Layer).

EDUCATION

Kalasalingam Academy of Research and Education, India| *Bachelor of Technology in Computer Science | CGPA: 8.95* | **2021 –2025**